AAYUSH KUMAR

Software Engineer | Backend (Python, FastAPI) | Full-Stack | AI / LLM

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SUMMARY

Software Engineer with a B.Tech from NIT Jaipur (2026) and production experience in backend, full-stack and AI engineering. In a backend internship, built async Python (FastAPI) REST APIs and RAG pipelines that serve LLM inference to 4,500+ users at 99.8% uptime, and cut API latency by 40%. As founder-developer of ARTHIS.space, scaled a 70+ game web platform, its React Native Android and iOS apps and a Python real-time multiplayer server to 10,000 daily active users on PostgreSQL (Supabase). Strong in Python, JavaScript/TypeScript, C++, data structures and algorithms, and system design.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, C#, Java, SQL, HTML, CSS

Backend: FastAPI (async), Node.js, REST API design, WebSocket, TCP sockets, PostgreSQL, Supabase, database design, Row Level Security, authentication, data pipelines, third-party API integration

AI / ML: LLM integration, RAG pipelines, vector databases, fine-tuning, transformer models (GPT-2), NLP, prompt engineering, AI agents, Model Context Protocol (MCP), Ollama

Frontend and Mobile: React, Vite, React Native (Expo), Android, iOS, Three.js, WebGL, HTML5 Canvas, responsive design

DevOps and Tools: Git, GitHub Actions, CI/CD, Linux, Docker, Cloudflare Workers, web security, performance optimization, technical documentation

CS Fundamentals: Data structures and algorithms, OOP, design patterns, system design, concurrency, multithreading, socket programming, software testing, debugging, Agile, SDLC

EXPERIENCE

Founder and Software Engineer | Arthis Technologies, Jaipur **Jun 2026 – Present**

arthis.space | arthis.land | github.com/itsAayush2004/arthisland-relay

- Built and scaled ARTHIS.space, a swipeable web platform of 70+ HTML5 single- and multiplayer games (Three.js, 2D Canvas, WebRTC), from 1 to 10,000 daily active users on Supabase and PostgreSQL.
- Designed a zero-dependency real-time multiplayer server (TCP and WebSocket) in pure Python, hosting 2–4 player rooms for Unity clients.
- Shipped Android and iOS apps in React Native (Expo) with a live game-preview feed and AdMob ads, optimized for low-end devices.
- Engineered a zero-cost 5-layer code-protection pipeline (obfuscation, AES-GCM encryption, Cloudflare Worker key service, WebAssembly VM) that secures 78 games.
- Built Arthis.Land, a React and Vite metaverse whose Unity WebGL viewer loads .glb rooms at runtime; automated CI/CD with GitHub Actions and wrote its architecture and security docs.
- Built a multi-agent AI game generator (build, score, fix loop) on Claude and Ollama, with RAG search and fine-tuning.

Backend Developer Intern (AI) | Internship, Jaipur **Jan 2026 – Jun 2026**

- Designed and shipped 6 production REST API endpoints in async Python (FastAPI), serving LLM inference to 4,500+ users.
- Cut API latency by 40% by rebuilding data pipelines and adding concurrent request handling, at 9,000 peak daily requests.
- Built RAG pipelines on vector databases to ground LLM answers in internal data; trained a GPT-2 model from scratch.
- Tested, deployed and monitored AI services on Linux in a 4-engineer Agile team.

Freelance 3D Animator and Tools Developer | AKVerse, Jaipur **Dec 2024 – Jan 2026**

- Developed 28 Blender Python add-ons (lip-sync, camera rigs, IK posing, render automation), replacing manual animation.
- Automated YouTube Shorts production (Claude API, edge-tts, Blender, FFmpeg) for AKVerse, a 21,000+ subscriber channel.

PROJECTS

- **CV Forge** (JavaScript, Node.js, Playwright): ATS CV tailoring app: extracts must-have keywords from a job post, scores CV match 0–100 and exports PDF and Word; tested on 25 real job posts. github.com/itsAayush2004/cv-forge
- **3D Portfolio and TravelAtlas** (Three.js, WebGL): Built a scroll-driven 3D gallery portfolio that doubles as a CV, and an interactive 3D globe charting 30+ travelled places. itsaayush2004.github.io/portfolio
- **HexaBed** (Unity, C#): Built a procedural hex-grid terrain generator with GLB asset placement that removes manual level layout.

EDUCATION

B.Tech, Electronics and Communication Engineering | Malaviya National Institute of Technology (MNIT, NIT Jaipur) **2022 – 2026**

JEE Advanced: AIR 8956 | JEE Main: 98.9 percentile | Class XII: 86.7% | Class X: 90.4%